

1. Administrative & Study Identification

Full Study Title: A Multicenter, Open-Label Phase IV Study to Evaluate Overall Health, Physical Activity, and Joint Outcomes, in Participants Aged ≥ 13 and < 70 Years With Severe or Moderate Hemophilia A Without FVIII Inhibitors on Emicizumab Prophylaxis **Short Title/Acronym:** MO42623 Study (Emicizumab Joint Outcomes Study) **Protocol Number:** MO42623 **NCT Number:** NCT05181618 **Sponsor/Company Name:** Roche **Investigational Product:** Emicizumab (Hemlibra) **Phase of Development:** Phase IV (PHASE4) **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Roche Clinical Trial Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the impact of emicizumab prophylaxis on joint outcomes by assessing joint status at 6, 12, and 24 months based on centrally reviewed Haemophilia Early Arthropathy Detection with Ultrasound (HEAD-US) scores, with a specific focus on the synovitis score in participants with synovitis. **Secondary Objectives:** To evaluate overall health, physical activity, and the safety and tolerability of emicizumab prophylaxis, specifically monitoring for adverse events (AEs), thromboembolic events, and events of thrombotic microangiopathy (TMA).

2.2 Background & Rationale To evaluate the real-world effectiveness and joint health improvements in participants aged ≥ 13 and < 70 years with severe or moderate hemophilia A (without FVIII inhibitors) who are transitioning from standard FVIII prophylaxis to emicizumab prophylaxis. Understanding the long-term impact on physical activity and arthropathy (joint structural damage) is critical for optimizing hemophilia care.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Three-cohort open-label study) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 136 evaluable patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** [Study Start Date] **Estimated Primary Completion:** [24 Months post-enrollment]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 13 and < 70 years old. 2. Diagnosis of severe congenital hemophilia A (intrinsic FVIII level $< 1\%$) or moderate congenital hemophilia A (intrinsic FVIII level $\leq 5\%$) previously prescribed prophylaxis for at least the last 24 weeks. 3. Negative test for FVIII inhibitor (< 0.6 Bethesda Units) during the screening period and no history of FVIII inhibitory antibodies in the last 5 years.

4.2 Exclusion Criteria 1. Previous treatment with emicizumab prophylaxis. 2. Planned joint replacement, joint procedure, or synovectomy at screening, or conditions other than hemophilia A that can affect joint health (e.g., osteoarthritis). 3. Pre-existing uncontrolled cardiovascular disease, history of illicit drug/alcohol abuse, or high risk for thrombotic microangiopathy (TMA) and thromboembolic disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen **Drug Name:** Emicizumab **Trade Name:** Hemlibra **ATC Code:** B02BX06 **Dosage/Frequency:** Emicizumab prophylaxis (loading dose phase followed by regular maintenance doses). **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** Up to 24 months of study evaluation. **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/hemlibra-epar-product-information_en.pdf

5.2 Concomitant & Prohibited Therapy Permitted: Standard on-demand treatments for breakthrough bleeds (per protocol guidelines). **Prohibited:** Investigational drugs to treat/reduce hemophilic bleeds within 5 half-lives at screening; medications known to unacceptably increase the risk of thrombosis.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, FVIII Inhibitor Lab Testing, MRI eligibility check
Baseline	Day 0	Emicizumab Loading Dose Initiation, Baseline HEAD-US Ultrasound
Interim 1	Month 6	Safety Labs, Efficacy Assessment, HEAD-US Joint Status Evaluation
Interim 2	Month 12	Safety Labs, Physical Activity Tracking, HEAD-US Joint Status Evaluation
End of Study	Month 24	Final HEAD-US Evaluation, Exit Interview, Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Joint Status at 6 Months, 12 Months, and 24 Months focusing on the Synovitis Score in

participants with synovitis. **Measurement Method:** Centrally Reviewed Haemophilia Early Arthropathy Detection with Ultrasound (HEAD-US) Scores.

7.2 Secondary & Exploratory Endpoints 1. Number of participants with at least one Adverse Event (Severity determined by WHO Toxicity Scale). 2. Number of participants with at least one Thromboembolic Event. 3. Number of participants with at least one event of Thrombotic Microangiopathy (TMA).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine clinic visits, patient diaries for bleeding episodes, and laboratory monitoring. **Serious Adverse Events (SAEs):** Specific monitoring pathways for thromboembolic events and TMA; reported within 24 hours. **Data Monitoring Committee (DMC):** Internal/External safety review board overseeing Phase IV post-market continuous safety events.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for joint outcome efficacy; Safety Set for AE/SAE, TMA, and thromboembolic event rates. **Sample Size Justification:** Target $N=136$ adequately powered to detect changes in HEAD-US synovitis scores and track real-world physical activity metrics. **Handling of Missing Data:** Mixed-effects Model for Repeated Measures (MMRM) or Last Observation Carried Forward (LOCF) for ultrasound joint scores.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite monitoring, central review of all ultrasound (HEAD-US) and MRI images to ensure standardized scoring. **Audit Trail:** Secure electronic Data Capture (eDC) system with eCRF locking and query resolution workflows.

11. Study Identifiers & Governance

Posting ID / Primary ID: 630308152394493890 (Internal / Protocol: M042623) **Registry Number:** NCT05181618 **Sponsor/Lead Organization:** Roche **Study Title:** A Study to Evaluate Overall Health, Physical Activity, and Joint Outcomes in Participants With Severe or Moderate Hemophilia A Without Factor VIII Inhibitors on Emicizumab Prophylaxis **Official Regulatory Title:** A Multicenter, Open-Label Phase IV Study to Evaluate Overall Health, Physical Activity, and Joint Outcomes, in Participants Aged ≥ 13 and

\$<70\$ Years With Severe or Moderate Hemophilia A Without FVIII Inhibitors on Emicizumab Prophylaxis **Source Level:** 5

12. Clinical Framework (Hemophilia A Specific)

Name / Preferred Name: Hemophilia A **Preferred UMLS Name:** Hemophilia A without Factor VIII inhibitor **Semantic Type:** Disease or Syndrome **Disease Areas:** Severe Hemophilia A, Moderate Hemophilia A **Definition:** The classic hemophilia resulting from a deficiency of factor VIII. It is an inherited disorder of blood coagulation characterized by a permanent tendency to hemorrhage. **Diagnosis Threshold:** Intrinsic FVIII level \$<1\%\$ (Severe) or \$\\le 5\%\$ (Moderate) **MeSH Code:** D006467 **SNOMED ID:** 28293008 **Medical Methodology:** Transition from FVIII replacement to non-factor prophylaxis (Emicizumab) **Optimization Target:** Prevention of bleeding and improvement/stabilization of joint structural health (Synovitis)

13. Study Procedures & Methodology

Management Pathway: Standard clinical transition pathway for Emicizumab prophylaxis **Education Protocol (HCP):** HEAD-US scoring calibration and standardized administration of subcutaneous prophylaxis **Education Protocol (Patient):** Self-administration training for subcutaneous emicizumab, physical activity tracking guidance **Quality Audit Mechanism:** Centralized imaging review for joint status to eliminate local reader bias

14. Observation & Follow-Up Schedule

Total Duration: 24 Months (104 weeks) follow-up **Onsite Visit Cadence:** Months 6, 12, 18, and 24 for imaging and deep clinical assessment **Remote Monitoring Frequency:** Routine telemedicine/diary checks for bleeding events **Checklist Review Interval:** Continuous bleed tracking and periodic AE review

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				